

The economic feasibility of digital health and telerehabilitation

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Abstract

Information technology advancements have increased the capacity to include patients in the healthcare process, promote changes in health behavior, and provide the opportunity to widely disseminate lifestyle self-management programs. The effects of digital exclusion are widespread and will have an influence on people's access to services, employment markets, and social and civic involvement because technology is at the heart of current economic and social endeavors. Digital technology has the ability to enhance societal and personal health and welfare. The main purposes of digital health technology include wearable activity monitors that promote self-monitoring of physical activity, sleep, and heart rate, enable goal-setting, and deliver motivation for behavior change. Standard methodological principles for the assessment of health care technology, such as pharmaceutical medications and medical devices, are used in the majority of published economic evaluations of Digital Health Interventions. Impacts are expected to be more varied since digital health interventions costs and benefits frequently depend on how the intervention develops over time and how much the user interacts with it. The change in important product components and its immediate effects on resource consumption, benefit measurement, and assessment must be included in the economic evaluation. Quality-adjusted life years, a common metric in cost utility assessments, combines the health impacts in terms of both quantity (life years) and health-related quality of life. The World Health Organization (WHO) makes an explicit recommendation to use digital tools for healthcare professional education and training so that they can supplement rather than replace current educational initiatives. The WHO claims that this new distribution method can increase access to health education more affordably. It is important to address the possibility that developing country healthcare ecosystems are not yet developed enough to incorporate and integrate digital tools for healthcare. Americans spent around 18% of their gross domestic product on health care in the previous year. Although it is dramatic, other nations are also seeing increased cost pressures from health spending as their population ages. The economic viability of a Digital Health Intervention decides if its users, administrators, and developers are satisfied with it.

Keywords: Delivery of health care; Digital health; E-Health; Economic feasibility; Health profession; Mobile health; Remote patient monitoring; Telerehabilitation.

1. Introduction

Particularly in the developing world, technological advancements in a variety of fields, including the Internet of Things (IoT), big data analytics, artificial intelligence (AI), augmented reality, blockchain technology, and mobile technologies, are expected to have an impact on how healthcare and other services are provided (Chandwani et al., 2018). Digital healthcare is described as “the transfer of health resources and healthcare by electronic means” by the World Health Organization (WHO). The term “digital health” (also known as “e-health”) refers to health services and information that are provided through the Internet and other similar technologies. It is an emerging topic in the fields of medical informatics, public health, and commerce (Naoum et al., 2021). The ability to implement a program or intervention was characterized as feasibility, which was assessed in terms of acceptability, adherence, the probability that it would be cost-effective, or the availability of providers to carry out the intervention (Moschonis et al., 2023). By reducing the time between the onset of a symptom and its reporting, digital health interventions are utilized in palliative care to close the communication gap between medical practitioners and patients with life-threatening illnesses. The objective of the economic assessment of digital health treatments is to inform decision-makers about their relative value for money when compared to certain alternatives (Monitoring and Evaluating Digital Health Interventions: A Practical Guide to Conducting Research and Assessment, 2016). The 2030 Agenda for Sustainable Development emphasizes that there is significant potential for human development to be accelerated, the digital gap to be closed, and knowledge societies to be built via the expansion of information and communications technology (Miller et al., 2020). Although the IoT, virtual care, remote monitoring, AI, big data analytics, blockchain, smart wearables, platforms, tools enabling data exchange and storage, tools enabling remote data capture, the exchange of data, and the sharing of relevant information across the health ecosystem creating a continuum of care have proven potential to improve health outcomes (Rao et al., 2020). The branch of medicine known as digital health is one that is fast developing. It has a big impact on clinical research, patient care, and health care quality and efficacy (Zhang & Zhang, 2016).

Telemedicine is used to treat everything from health issues like myocardial infarctions (Pearson & Jordan, 2010) to psychiatric illnesses like autism (Mohanta et al., 2019). Telemedicine is an excellent option for follow-ups since it may lower the expense of housing patients in hospitals and enable them to see a doctor even in remote locations (Gentili et al., 2022). Traumas and joint arthroplasty are two instances of telemedicine use (Clarke et al., 2018). Health care practitioners may analyze, diagnose, monitor, treat, and educate patients “virtually” with the use of a variety of technology, such as “smartphones,” computer tablets, mobile apps, and video conferencing (Whetten et al., 2018). The fastest-

growing area of the healthcare industry right now is telemedicine. The delivery of medical treatments remotely is theoretically possible via telemedicine. Via telemedicine, distant nations and regions have begun to connect more deeply. Wide-scale change is difficult to achieve without a socially organized people (Kyratsis et al., 2022), access to electricity, and political organization. These problems are systemic in nature. It is simple to assume that a technology has failed when in reality it is solely not always applicable to the setting of developing nations (Demaerschalk et al., 2013). Finding a match between the demands of a setting and the healthcare model provided, or suitable healthcare, makes a significant difference in the outcomes in terms of life and death (Mosnaim et al., 2020). The absence of data on rural people makes it considerably more difficult to make evidence-based public health choices, which is a key obstacle to providing good treatment in developing countries (Lawrence, 2022, pp. 121–130).

2. Factor responsible for feasibility of digital health and telerehabilitation

2.1 Cost effectiveness

Cost effectiveness is a crucial factor in determining the feasibility of digital health solutions. Implementing digital health solutions may involve upfront costs such as developing or acquiring software, hardware infrastructure, training healthcare professionals, and integrating systems. The feasibility of digital health depends on whether these costs can be justified by the potential benefits and cost savings over time (Global Strategy on Digital Health, n.d.). The main issues in healthcare 4.0 include seamless data transfer rates with little to no data loss, traffic-free transmission channels, cost-effective, instantaneous data retrieval, and machine to machine or device to device connectivity in the IoT era (Digital Technologies: A New Determinant of Health, 2021). Digital health solutions often require ongoing operational expenses, including maintenance, technical support, data storage, and security measures. Many facets of eHealth are covered by digital technology, including telemedicine, mobile health, and health information systems. The advancement of disciplines like AI, big data analytics, and genomics improves eHealth as well (Bouabida et al., 2021). Evaluating the long-term operational costs is necessary to assess the overall cost effectiveness. The feasibility of digital health can be enhanced by its scalability (Greenhalgh et al., 2021). Solutions that can be easily scaled and deployed across a large number of users or healthcare settings can leverage economies of scale, making them more cost effective. Technology may minimize negative effects on patient populations that are spread out geographically by remotely connecting them to skilled doctors; nevertheless, new technologies and programs need to be reviewed in order to comprehend their costs and results (Wan et al., 2019). The affordability of digital health solutions is an important factor in their feasibility, especially in resource-constrained

settings or for underserved populations. Addressing affordability barriers, such as providing subsidies or incentives, ensuring reimbursement mechanisms, and promoting accessibility, can enhance the cost effectiveness and feasibility of digital health interventions (Chang et al., 2013). Programs used in video conferencing systems enable remote patients with any illness or sickness to receive expert advice through video (Jo et al., 2019). The sustainability and cost-effectiveness of digital care should be discussed by legislators. Building public trust and reaffirming a commitment to protect their privacy are essential (Shei et al., 2022).

2.2 User adoption

User adoption plays an important role in evaluating feasibility of digital health. Human health might undergo a revolution if adoption of effective digital health technologies is accelerated and their integration into normal care procedures is advanced. This would increase efficiency, reduce costs, and expand access to and capacity for care delivery (Degroote et al., 2020). Digital health solutions have the potential to generate cost savings in various ways. For example, telemedicine can reduce the need for in-person visits, which can lower transportation costs for patients and decrease hospital or clinic operational expenses. Patients from rural hospitals should only be transferred if they need a higher level of treatment (Maher et al., 2017). The feasibility of digital health can be enhanced by its scalability. Solutions that can be easily scaled and deployed across a large number of users or healthcare settings can leverage economies of scale, making them more cost effective. It is also important to consider the potential costs of not implementing digital health solutions. The affordability of digital health solutions is a crucial factor in their feasibility, especially in resource-constrained settings or for underserved populations. Addressing affordability barriers, such as providing subsidies or incentives, ensuring reimbursement mechanisms, and promoting accessibility, can enhance the cost effectiveness and feasibility of digital health interventions (Ferguson et al., 2022). Continuous evaluation of digital health solutions is important to identify user feedback, preferences, and areas for improvement. Incorporating user feedback into iterative design and development processes ensures that the technology aligns with user needs and expectations, increasing the likelihood of adoption. In response to the healthcare crisis, more changes in the adoption and management of remote health care happened in the first 20 days of March 2020 than in the preceding 20 years combined due to pandemic COVID-19 (Horgan et al., 2020).

The idea of ensuring that every person has an equal chance to attain optimal health outcomes, regardless of their social, economic, or demographic attributes, is known as “**health equity**.” While some health differences can be attributed to individual differences in biology, physiology, or genetics, health disparities (Curtis et al., 2021) are caused by

social and structural factors that place an individual or population at a social disadvantage and negatively affect how they interact with healthcare on a daily basis. Disparities in health systematically place those who are already at a disadvantage in terms of their health at a disadvantage, having detrimental knock-on implications on their capacity to advance socially, politically, and economically (Haleem et al., 2021). Social determinants of health, or the circumstances under which people are born, grow, live, work, and age, have an impact on health inequities, which in turn are influenced by health disparities, which are “systematic differences in the health of groups and communities occupying unequal positions in society that are avoidable and unjust.” Digital health has the ability to close gaps in current healthcare and advance equity in healthcare. Digital health solutions may reach underprivileged groups, rural locations, and those with little access to conventional healthcare services by utilizing technology. But in order to achieve true health equity, it is imperative to remove the obstacles that can prevent fair digital health adoption. The WHO declared that “Digital health should be an integral part of health priorities and benefit people in a way that is ethical, safe, secure, reliable, equitable, and sustainable” in their global strategy on digital health for the years 2020–2025 (Funderskov et al., 2019). The complementary approaches of human-centered design and Agile software development, which are well established in the tech sector and are emerging as potent tools for product development in healthcare delivery and research, can help ensure equitable digital health product design and technical development. Additionally, healthcare policies and funding should prioritize the equitable distribution and implementation of digital health initiatives, aiming to reduce disparities and improve health outcomes for marginalized communities (Van Gemert-Pijnen et al., 2011). The wider healthcare system is a final important stakeholder in digital health since it influences particular use cases for these technologies (such as AI for radiology practices) and may help or impede their successful, widespread acceptance and implementation. The development of infrastructure for data integration, interoperability, and analysis in a way that promotes “data solidarity”—an approach to health data that emphasize data justice and equity and centers those potentially disadvantaged by health data and technology use—is one of the top priorities for healthcare systems to support digital health equity (Motivating Patients to Use Smartphone Health Apps, 2015).

3. Ways to deliver digital health

3.1 Remote patient monitoring

A tele neurology program not only yields financial rewards but also improves patient outcomes, contributes to society by reducing stroke-related impairment, and makes it more convenient for patients’ relatives. By eliminating the need for transportation to an urban tertiary care center, it significantly reduces the financial burden on rural hospitals

(Berwick, 2003). Planning and Evaluating Remote Consultation Services takes into account how these different areas interact and change over time as a complex system and has seven domains: the reason for consulting, the patient, the clinical relationship, the home and family, technologies, staff, the healthcare organization, and the wider system (Long et al., 2018). Remote patient monitoring involves the use of connected devices, such as wearable sensors, to collect and transmit patient health data to healthcare providers for remote monitoring. It enables continuous tracking of vital signs, activity levels, medication adherence, and other relevant health metrics. RPM allows healthcare providers to remotely monitor patients' health status, detect potential issues early on, and provide timely interventions. In a survey during COVID-19 pandemic for telecare platform, first the participants appreciated how simple it was to get services generally and, in particular, how readily available and reachable the care team was. The second was that they valued the clinical oversight and continuity of treatment from the hospital to the home (Muke et al., 2019). Telerehabilitation was discovered to be a secure, well-tolerated, and economical method of giving patients treatment at home (He et al., 2019).

3.2 Video conferencing

A more advantageous technology is telemedicine, which can facilitate people's access to preventative care and improve their long-term health. This is especially true for people who encounter barriers to accessing adequate care due to financial or geographic constraints. Tele-health has the potential to improve the efficiency, coordination, and accessibility of healthcare (Medhanyie et al., 2015). Young individuals with diabetes can get care using a safe and effective combination of home telemedicine and shared doctor visits. The concept promoted young adults' participation in treatment by improving clinic follow-up and patient appointment satisfaction (Donkin et al., 2011). For patients in rural places, telemedicine technology has immense potential. The biggest impact is seen in a number of nations where there are little or no healthcare services. Patients will not have to wait in long lines, and doctors will have easier, more effective access to patient data thanks to electronic records and shorter wait times. Additionally, by scheduling remote visits, medical professionals may serve a larger number of patients in less time. Telemedicine equipment communication applications have changed and become more immersive due to virtual reality (VR) technologies. In VR, surgeons and their colleagues may now execute surgical procedures while watching a 3D display. Doctors and even surgeons may operate on patients who are thousands of kilometers distant by using video conferencing (Nadarzynski et al., 2019).

3.3 Mobile health (mHealth)

The use of mobile and wireless technology for health is known as “mobile health” (mHealth), and it has the potential to enhance health outcomes by addressing important health system limits to service delivery, coverage, and utilization. For a wide range of medical illnesses, text message-based health treatments provide patients reminders, information, or help with self-management. mHealth tactics of SMS reminders for treatment adherence for HIV/ART treatment, malaria and TB therapies, and smoking cessation are included in the majority of extant economic assessment studies. With certain processing speed and quality improvement measures, several research studies employed mHealth techniques for data collecting, family planning education, and telephone assistance for breastfeeding (Jarva et al., 2022). A pregnancy monitoring system based on a mobile phone may be cost effectively expanded to include SMS and home visit reminders (Morris et al., 2018). mHealth offers several advantages, such as enhanced accessibility, convenience, and personalized care. It allows individuals to take an active role in managing their health, facilitating self-monitoring and self-care. Mobile health apps can also support healthcare providers by enabling remote patient monitoring, facilitating communication and data exchange, and improving care coordination. However, challenges such as ensuring data privacy and security, addressing the digital divide, and maintaining the quality and reliability of mobile health apps need to be addressed for wider adoption and effective implementation.

3.4 Wearable activity monitor

Wearable activity monitors are an essential component of digital health, providing individuals with real-time data and insights into their physical activity levels and overall health. These small, portable devices, typically worn on the wrist or clipped to clothing, track and measure various metrics, including steps taken, distance traveled, calories burned, heart rate (HR), and sleep patterns. Elite, competitive, and recreational athletes, as well as both the physically active and sedentary general populace, have become more and more interested in wearable technology (Porterfield et al., 2014). Among wearable devices are digital 3-axis accelerometers, magnetometers, thermometers, optical HR sensors, and 3-axis pulse oximeters. Fitness companies are marketing these devices as being able to estimate and monitor physiological parameters like step count, HR, sleep quality, sleep rhythm, energy expenditure, maximal oxygen consumption ($\dot{V} O_2 \text{ max}$), peripheral oxygen saturation, and the “training effect” of both individual and cumulative exercise bouts thanks to these sensors and ever-improving algorithms—most of them proprietary

(Skovlund et al., 2019). Wearable activity monitors also contribute to preventive care by increasing awareness and self-monitoring. Users can gain insights into their sedentary behavior and make conscious efforts to incorporate more movement into their daily routines. Despite their benefits, challenges remain in ensuring the accuracy and reliability of wearable activity monitors. It is crucial to verify the quality and validity of the data collected by these devices, as inaccuracies may lead to misguided conclusions or inappropriate actions. Moreover, issues related to data privacy and security must be carefully addressed to protect users' sensitive health information. These gadgets promote lifestyle physical activity, or the accumulation of physical activity via daily activities, and are inexpensive (Keller et al., 2015), pleasing to the eye, and user-friendly. As a result, they lower the obstacles associated with more regimented types of physical exercise. Activity trackers are favorably received by users and are utilized for a sizable amount of time. Most respondents said that using the activity tracker had enhanced their level of physical activity. Breakage or loss of the device and technical issues with the device and supporting software were major obstacles to ongoing usage (WHO Guideline Recommendations on Digital Interventions for Health System Strengthening, n.d.).

3.5 Virtual assistant

Virtual assistants are increasingly being utilized in the field of digital health to provide personalized support, information, and guidance to individuals. These intelligent virtual assistants, powered by AI, can interact with users through voice or text interfaces, offering a range of healthcare services and resources. The future of automated, personalized, easy health support holds promises thanks to advancements in machine learning and AI, which is already transforming the way that health care is provided (Vayena et al., 2013). Essentially, virtual assistants are digital services created to mimic human communication and offer tailored replies based on input from the user. They can be designed to have predetermined discussions or to respond to queries from the user. From basic menu- or multiple choice—based virtual assistants to more advanced ones with natural language processing that can understand free speech or text, capabilities differ widely (Greaves et al., 2018). These AI-powered assistants may help with a variety of tasks, including symptom assessment and triage, medication reminders, advice on choosing a healthy lifestyle, and answering health-related concerns. Virtual assistants have the potential to enhance healthcare provider workflows by automating routine tasks, facilitating appointment scheduling, and providing real-time access to patient information. Nowadays AI powered imaging for better 3D medical scans and advanced radiographs are available. Virtual nursing assistants are there to ease out the medical services. Virtual clinics have become commonplace in order to maintain long-term patient care and to continue seeing new referrals. To make a diagnosis, several of them turned to innovative remote testing technologies (Manteghinejad & Javanmard, 2021).

4. Acceptability of digital health among care workers

Central to digital health research and practice is the degree to which interventions are viewed as “acceptable” by patients, families, healthcare professionals, institutional review boards, and policy makers (O’Connor et al., 2017). Because technology does not appear to be working for them or the benefit of their patients, health care professionals are frequently apprehensive and offer minimal support for e-Health. As a result, acceptance issues with e-Health technology are common. We are aware that for digital health initiatives to be successful at the individual level, individuals must actively participate in them. Unfortunately, engagement levels are frequently below average. Poor acceptability may, at least in part, be responsible for this poor participation (Panch et al., 2019).

4.1 Positive reinforcement of digital health

The effectiveness of digital health technologies is especially valued by healthcare professionals who operate in rural and distant areas since they enable them to provide services through the device. Because they allow them to work more quickly, digital health technologies are likely to be perceived by health professionals as being more efficient. It has been suggested that those who seek to accelerate the adoption of new technologies within their organization should invest in “early adopters” (i.e., trendsetters who differ from the majority in that they are not dependent on social proofing) and make their activities visible in an effort to alter prevailing social and cultural norms, which are deeply contextual and constantly changing (Cummins & Schuller, 2020). Additionally, these technologies are anticipated to save travel time for health professionals in both urban and rural settings, allowing them to spend more time with their clients in the former or give remote assistance to clients in the latter. In terms of possible time and money savings, ease, and the ability to learn at their own speed, the digital platform seems helpful for supporting the training of social health advocates (Ronquillo et al., 2023). The portability of digital health technology may be appreciated by healthcare professionals as it allows them to be flexible, work when it is convenient, and access information without being confined to an office. Furthermore, some health professionals claim that the use of digital health technology has improved their status in society and increased the level of community trust and respect they get. AI has been used in medicine and a number of healthcare services, including genetic diagnostics, clinical laboratory testing, and health communications, as well as diagnostic imaging, among others. These systems provide useful medical data to doctors, notifications regarding any high-risk health consequences, and diagnostic or treatment error reduction (O’Malley et al., 2022).

4.2 Negative reinforcement of digital health

Healthcare providers may not find digital health treatments to be as effective since they do not see a reduction in their workload, and in other cases, an increase, making them less

willing to adopt such interventions. Other digital intervention studies have also noted technical issues like inadequate internet access, unresponsive device displays, and slower device performance (Butcher & Hussain, 2022). We are aware that for digital health initiatives to be successful at the individual level, individuals must actively participate in them. Unfortunately, engagement levels are frequently below average. Poor acceptability may, at least in part, be responsible for this poor participation (Kasia, n.d.). New technologies are frequently introduced to the market and used as part of digital health interventions, such as chatbots and wearable devices. According to research, while internet users in 2019 saw chatbots as somewhat acceptable, there remained concern surrounding the quality, accuracy, and reliability of their information (Grunwez et al., 2021). Health professionals may also be worried about theft, loss, and damage. They may also object to having to carry both a personal and a business phone. According to the healthcare professionals that were interviewed using information technology and diverse digital health systems reported that digital health involves a variety of skills, such as expertise with various information and communication technologies and problem-solving techniques when faced with technical issues (Piwek et al., 2016). Health professionals who find it difficult to utilize these technologies may not grasp the information they provide, have a poor opinion of their value, and worry about making mistakes. In some cases, a lack of computer literacy puts at risk employment stability. However, even if a service is straightforward and simple to use, a healthcare professional may occasionally need to troubleshoot to resolve technical difficulties, which calls for some IT expertise (Yang & Hsu, 2010). One of the primary concerns about chatbots is that they lack empathy, namely the ability to understand users' emotional states and customize answers to suit those circumstances. Therefore, the interaction with health chatbots may be hampered by the lack of empathy (Swan, 2009).

5. Feasibility of digital health among patients

E-prescribing, in which prescriptions are transmitted to local pharmacies electronically, is on the rise, but information systems rarely interact with one another to share data on patient safety and billing (Swan, 2012). On the other hand, people who deal with illnesses that are frequently stigmatized or highly personal (such as HIV, family planning, and abortion care) worry that their participation in targeted communication program will lead to the disclosure of their private health information or the tracking down of their identities. Clinical practice may benefit from using digital patient-reported outcome technologies to enhance parts of person-centered chronic disease treatment and possible outcomes of care (Vogt & Kuperberg, 1996). Some people who use telemedicine services prefer in-person interactions. Sharing electronic health record access should facilitate more effective communication while minimizing disruptions to the doctor's office's routine (Benjamin et al., 2018). For some people, getting access to and using these services might be very

challenging. These include clients who speak minority languages, have little reading or digital literacy abilities, or have hearing impairments; people with minimal access to network services, energy, or mobile devices, patients with health issues that are disregarded (US Food & Drug Administration & Center for Devices & Radiological Health Digital Health Program. Digital Health Innovation Action Plan (United States Food and Drug Administration, 2018)). The ethical issues with health digitization are equally significant. One of these ethical issues is user permission. Users should be informed about data gathering. Almost of users merely click the “I agree” button at first without reading the conditions of use of the programs, even though most applications ask for this permission (Goodman & Flaxman, 2017). The absence of information regarding how such strategies may affect health outcomes, health system efficiency, and the cost-effectiveness of service delivery is the primary knowledge gap regarding the adoption of digital health techniques (Sussillo & Barak, 2013). Security and privacy issues arise when data are transferred from the site of collection, such as IoT devices, to remote servers, and these issues must be resolved. Although informed consent and privacy by design are well-established practices in digital health, there are still issues about patients’ comprehension of how and by whom their data are used (Kruse et al., 2016). Any AI-driven clinical or policy intervention needs the confidence of the people who give their data and whose lives are touched, as well as the trust of doctors and policy officials who may utilize AI to guide choices, in order to have a beneficial impact (Hashimoto et al., 2018).

6. Discussion

Digital technology has the ability to enhance communal and domestic health and welfare. Digital health encompasses the use of technology and digital platforms to improve healthcare access, monitoring, and delivery, while telerehabilitation specifically focuses on remote rehabilitation services (Patrzyk et al., 2017). The integration of these two fields has the potential to transform the way rehabilitation is provided, benefiting patients, healthcare providers, and the healthcare system as a whole. When it is not feasible to wait for lifetime data to evaluate the cost-effectiveness, modeling approaches are typically employed to forecast the impact and possible cost (or savings) of a chosen technology (Moya et al., 2010). The sort of digital health intervention that was most often used “videoconferencing system” also had the most inconsistent outcomes (Handscho et al., 2014). The data supporting the cost-effectiveness of digital technologies in the health industry are still scant and restricted, despite the increased interest in investing in them. In a wide range of clinical and nonclinical populations and age ranges, wearable activity-tracker therapies are helpful in boosting physical activity and assisting with moderate weight reduction (Michaud et al., 2018). Ensuring an application does not negatively affect the performance of other programs running on a given device is one of its main objectives. As technology continues to advance and the acceptance and integration of

telerehabilitation grow, it is expected to become an integral part of comprehensive rehabilitation care, benefiting individuals in need of rehabilitation support (Bradford et al., 2014; Iribarren et al., 2017). Increasingly, it is being discovered that using digital technology may help low- and middle-income nations improve the monitoring and training of community health workers as well as the delivery of healthcare (van der Hout et al., 2021). By analyzing the vast volumes of patient data generated by high-tech gadgets from many sources, digital health will help meet the medical field's future demands (Lau & Kuziemy, 2017). AI may make use of the data produced by digital health systems to aid in areas of medicine including enhanced diagnosis, therapy selection, and clinical outcome prediction. The existence of AI solutions in digital health exacerbates issues with fairness, safety, and explanation (Bergmo, 2017). The five essential challenges faced in digital health are societal factors, ethical challenges, more connected health solutions, AI, and potential of genomics (Finucane et al., 2021). Digital health is gaining popularity since it will allegedly: increasing healthcare access, the healthcare system's inefficiencies should be reduced, boost the level of care provided, reduce the price of healthcare, more individualized patient care should be provided (Hamlin et al., 2019; Li et al., 2017).

7. Conclusion

The economic feasibility of digital health and telerehabilitation is dependent on various factors like cost effectiveness, user adoption, and health equity. Methods to deliver digital health include mobile health like text messages, mobile-based applications, etc, wearable devices, video conferencing, remote patient monitoring, and virtual assistant. Recent advances are emerging and improving the healthcare system with digital technologies. The combination of digital health and telerehabilitation holds great promise for revolutionizing rehabilitation services. For a digital health intervention to be successful, one should focus on effectiveness, acceptability, and feasibility. By enabling substantial changes in medical treatment both during the pandemic and thereafter, digital health technologies will play a crucial role in battling pandemic and prospective ones in the future.

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